A Unified Novel Architectural Paradigm for Remote Sensing Change Detection

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***Abstract*—In recent years, remote sensing change detection (CD) has garnered significant attention due to its critical role in resource monitoring and disaster assessment. CD tasks exist with different output granularities such as binary CD (BCD), semantic CD (SCD), and building damage assessment (BDA). However, existing methods require substantial expert knowledge to design specialized decoders that compensate for information loss during encoding across different tasks. This not only introduces uncertainty into the process of selecting optimal models for abrupt change scenarios (such as disaster outbreaks) but also limits the universality of these architectures. To address these challenges, a unified, general CD framework named unified novel architectural paradigm for remote sensing CD (UniRSCD) is proposed here. Building upon a state space model (SSM) backbone, a frequency change prompt generator (FCPG) as a unified encoder is introduced. The encoder dynamically scans bitemporal global context information while integrating high-frequency details with low-frequency holistic information, thereby eliminating the need for specialized decoders for feature compensation. Subsequently, the unified decoder and prediction head establish a shared representation space through hierarchical feature interaction and task-adaptive output mapping. This integrates various tasks such as BCD and SCD into a unified architecture, thereby accommodating the differing output granularity requirements of distinct CD tasks. Experimental results demonstrate that the proposed architecture can adapt to multiple CD tasks and achieves leading performance on five datasets, including the binary change dataset LEVIR-CD, the semantic change dataset SECOND, and the BDA dataset xBD. The source code is available at: https://github.com/zhenliuzhenqi**

***Index Terms*—Frequency awareness, remote sensing change detection (CD), state space model (SSM), unified framework.**

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I. Introduction

In RECENT years, remote sensing change detection (CD) has garnered widespread attention due to its crucial role in resource monitoring, environmental assessment, and disaster response. With the advancement of deep learning, various methods have been proposed to address diverse application needs, such as binary CD (BCD) [1], [7], [8], semantic CD (SCD) [2], [9], [10], and time-series analysis [3], [11].

In the field of remote sensing CD, inherent differences exist in feature interaction patterns and output granularity across various tasks, such as BCD [8], [9], [12], SCD [13], [14], [15], and building damage assessment (BDA) [3], [8], [16]. As shown in Fig. 1, to follow the task-driven design paradigm, existing methods require custom decoders [17] to compensate for information lost during encoding, as well as distinct prediction heads to adapt to different tasks. These methods are built upon two primary neural network architectures: convolutional neural networks (CNNs) [18], [19], [20] and visual transformers [21]. CNN-based approaches effectively capture local spatial features through convolutional inductive bias [22], [23], yet their limited receptive fields restrict modeling of long-range dependencies. On the contrary, transformers utilize self-attention to capture global information [8], [24], [25], but their uniform weighting often smooths out fine-grained details, reducing the model's focus on granular information. Therefore, existing methods employ customized decoders to compensate for information loss during the encoding phase, thereby addressing the aforementioned issues. However, this custom design hinders the development of a unified framework, thereby preventing the consolidation of diverse CD tasks with varying output granularities under a unified architecture.

To address the above problems, a generalized CD framework with a single-stream architecture, a unified novel architectural paradigm for remote sensing CD (UniRSCD) is proposed. This framework constructs its input by horizontally concatenating bitemporal images and utilizes state space models (SSMs) to capture long-range spatiotemporal dependencies [26] from shallow to deep layers. Meanwhile, UniRSCD introduces a frequency change prompt generator

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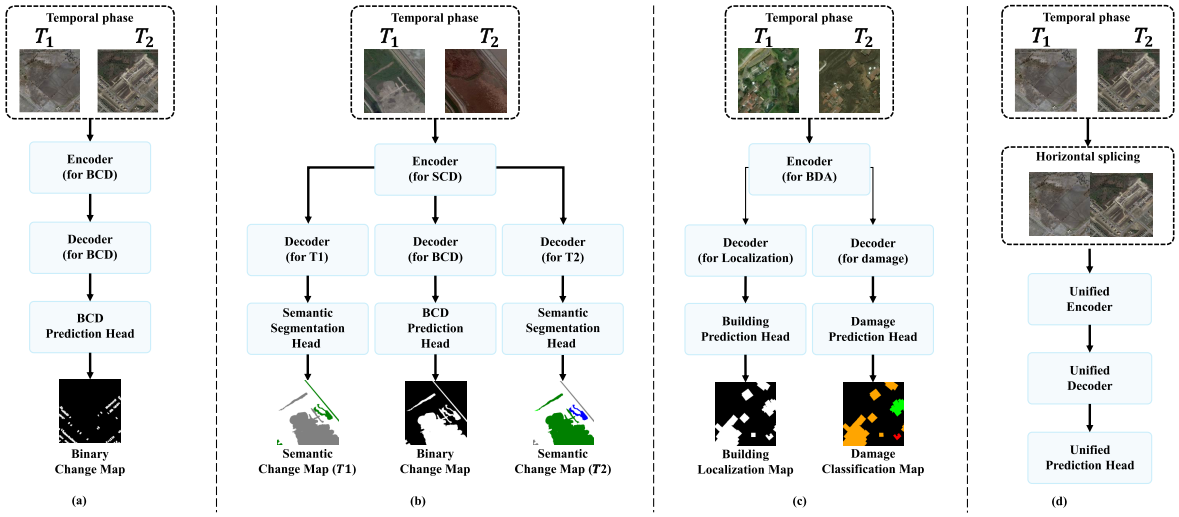


Fig. 1. Overall architectures for BCD task, SCD task, BDA task, and UniRSCD. (a) BCD task extracts features to identify whether surface regions have changed between bitemporal images and generates a binary change mask. (b) SCD task extracts features to locate semantic changes. (c) BDA task extracts features to perform both building localization and multilevel damage severity classification. (d) Proposed UniRSCD uses horizontal concatenation to process bitemporal images, then handles multiple tasks through a unified architecture.

(FCPG) module that implicitly formulates CD as a two-stage optimization process: first extracting low-frequency structures to locate candidate change regions, then integrating high-frequency details to refine boundaries. The design combining SSMs with frequency prompts compensates for information loss during encoding in existing methods, eliminating the need to build task-specific decoders for different CD tasks. In addition, the framework also designs a unified decoder and prediction head, which unifies multiple tasks into a shared representation space through hierarchical feature interaction and task-adaptive output mapping, thereby constructing a unified CD framework.

Experiments demonstrate that UniRSCD achieves an *IoU* [27] score of 72.03 on the SYSU BCD dataset, an *mIoU* score of 72.7 on the SECOND SCD dataset, and an $F_1^{overall}$ score of 78.02 on the BDA dataset. Benefiting from its single-stream architecture and simplified decoder design, UniRSCD significantly reduces training time compared to existing methods, enabling rapid and precise detection of sudden change scenarios. In summary, the main innovations of this article are reflected in the following three aspects.

1) We integrate an SSM with a FCPG module to effectively capture both global contextual information and local detailed features, thereby eliminating the need for specialized decoders to compensate for information loss during encoding.
2) We design a unified decoder and a multitask adaptive prediction head to unify various CD tasks.
3) We conduct experiments on three types of tasks (BCD, SCD, and BDA). The framework can quickly adapt to different tasks and achieves state-of-the-art (SOTA) results on the datasets.

II. Related Work

A. Deep Learning CD

Current CD communities have developed various architectures for different CD tasks. For example, it has branched into BCD for identifying surface changes [1], [7], [11], [28], SCD for localizing and categorizing changes [2], [10], [24], and BDA for classifying building damage levels postdisaster [3], [17]. Current mainstream CD approaches primarily rely on CNNs [18], [19], [20], [22], [29], [30], [31] and transformer architectures [21], [24], [32], [33], [34], [35]. CNNs excel at extracting local spatial features through their local connections and inductive biases [14], [23], [28], [36]. Conversely, visual transformers effectively capture global contextual information via self-attention mechanisms [24], [37], [38]. Although the above methods have developed different decoders and prediction heads for various tasks to compensate for information loss during the encoding process achieving high accuracy, this approach hinders the development of a unified architecture and also limits the adaptability and deployment efficiency of the model. We propose a unified encoding architecture that integrates the SSM with a FCPG module. This approach ensures both global consistency and local precision of features during the encoding phase, thereby eliminating the need for complex task-specific decoders. It enables high-performance detection across BCD, SCD, and BDA under one architecture, enhancing integration and scalability.

B. State Space Models

In recent years, SSMs have provided new perspectives for remote sensing CD due to their remarkable efficiency in long-sequence modeling. Particularly, architectures represented by Mamba [17], leveraging their linear time complexity and selective scanning mechanism, can efficiently capture global spatiotemporal dependencies in multitemporal remote sensing images, significantly enhancing the efficiency of processing

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high-resolution imagery and the capability of feature representation [17], [39], [40], [41], [42], [43], [44], [45], [46]. However, despite the significant progress these methods have achieved in computational performance and feature modeling, they generally lack specialized mechanisms designed to capture both global and local information in remote sensing CD, such as global contextual dependencies across temporal phases and fine-grained local details subject to environmental interference [4], [23], [47], [48]. Consequently, although existing SSM-based methods possess advantages in efficiency and feature extraction, their capability to integrate global and local information remains a key limitation and a core challenge to be addressed when dealing with complex real-world scenarios.

III. Method

A. Overview

Based on SSMs [17], [39], [40], we propose UniRSCD, a unified CD framework with a single-stream architecture [49] that enables multitask processing through a unified decoder and prediction head. The core design of this architecture integrates a frequency prompt mechanism into the SSM. This allows a single framework to handle multiple CD tasks without requiring dedicated decoders for each task. As shown in Fig. 2, UniRSCD comprises three modules: a unified change frequency prompt encoder (built upon an SSM backbone and introducing a FCPG), a unified change decoder, and a unified prediction head.

B. Unified Change Frequency Prompt Encoder

Taking concatenated bitemporal images horizontally as input, the unified change frequency prompt encoder consists of a four-stage backbone network based on SSMs, FCPGs integrated at each stage, and multiscale feature outputs, forming a unified encoding architecture specifically designed for remote sensing CD.

Horizontal Concatenation: Given the registered bitemporal images $X_{pre}, X_{post} \in R^{C\times H\times W}$, where C is the number of input channels, and H and W are the spatial dimensions, a horizontal concatenation is performed along the width dimension, as

$$X_{concat} = Concat_{width}\left(X_{pre}, X_{post}\right). \tag{1}$$

This concatenation better aligns with the 1-D sequence scanning mechanism of the SSM. Under typical image aspect ratios, stitching along the width allows the model to process corresponding row pixels in T_1 and T_2 more continuously during cross-temporal scanning. A four-directional cross-scanning strategy is employed to convert the 2-D feature map X_{concat} into sequential representations. In particular, this scanning method is executed in parallel by the underlying VSSM core operators. We scan along the row, column, and their reverse directions, flattening the 2-D features into a 1-D sequence $S_d \in R^{L\times C}$, where $L = H \times 2W$ denotes the sequence length. This multidirectional scanning captures cross-temporal features from diverse perspectives, offering comprehensive spatial context for CD. The flattened sequences are processed and then restored to spatial features via reverse scanning as $F_d = Reshape_d(S'_d)$, where S'_d denotes the sequence transformed by the visual state space module. Finally, features from all four directions are aggregated to form a unified multiscale representation.

Unified Change Frequency Prompt Encoder: The proposed change frequency prompt backbone network consists of an SSM backbone network and a FCPG.

1) SSM Backbone: We adopt the Selective SSM [26] as the core building block to model complex spatiotemporal dependencies in bitemporal features. Given an input sequence representation $x_t \in R^{L\times D}$, where L is the sequence length and D is the feature dimension, the SSM achieves information propagation and integration through the hidden state $h_t \in R^N$. Its continuous-time form is expressed as

$$\begin{aligned} y(t) &= Ch(t) + Dx(t) \\ h'(t) &= Ah(t) + Bx(t) \end{aligned} \tag{2}$$

the selective SSM employs input-dependent parameters for B, C, and Δ, which are generated from the input x_t via linear projections as $\Delta = softplus(W_{dt}x_t), B = W_B x_t, C = W_C x_t$, where W_{dt}, W_B, W_C are learnable weights. This adaptive parameterization enhances sensitivity to subtle changes by enabling the model to focus on relevant information while ignoring redundancies. The visual state space block integrates the state space branch and the MLP branch into a unified architecture. The state space branch uses layer normalization, a selective SSM, and deep convolutions to model global dependencies and local features. The MLP branch employs fully connected layers and *GELU* for nonlinear transformation. Outputs from both branches are fused with the inputs via stochastic depth, forming a residual learning framework that captures both global patterns and local interactions, thereby providing multiscale hierarchical features for CD.

2) Frequency Change Prompt Generator: Inspired by frequency-selective processing in human vision [33], [50], we design a FCPG to enhance change-related features in the frequency domain. For an input feature $X \in R^{C\times H\times W}$, the FCPG computes the radial frequency distribution $F_{radial} = (F_h^2 + F_w^2)^{1/2}$, where F_h and F_w are the frequency components along the height and width. Learnable thresholds are then used to construct low-frequency and high-frequency masks. The initial thresholds of 0.1 and 0.25 are empirical values used to separate high-frequency and low-frequency information. As shown in Fig. 2, each frequency band first passes through a global frequency prompt generator for base weights, followed by an adaptive attention mechanism for spatial refinement. The final prompt map for each band is obtained via a triple-product operation [26]

$$P_b = W_b^{global} \odot M_b \odot W_b^{spatial}. \tag{3}$$

The separated frequency band features are then reintegrated into the original feature space using a feature fusion convolution. To preserve spatial continuity, a spatial modulator (SPM) is applied to enhance local consistency, producing the modulated output as $P_{modulated} = P_{fused} \odot SPM(P_{grouped})$, where $P_{grouped}$ represents the mean features after channel grouping. The modulated frequency prompts are fused with the original features via a residual connection, as $X_{output} = X + \alpha \cdot P_{modulated}$,

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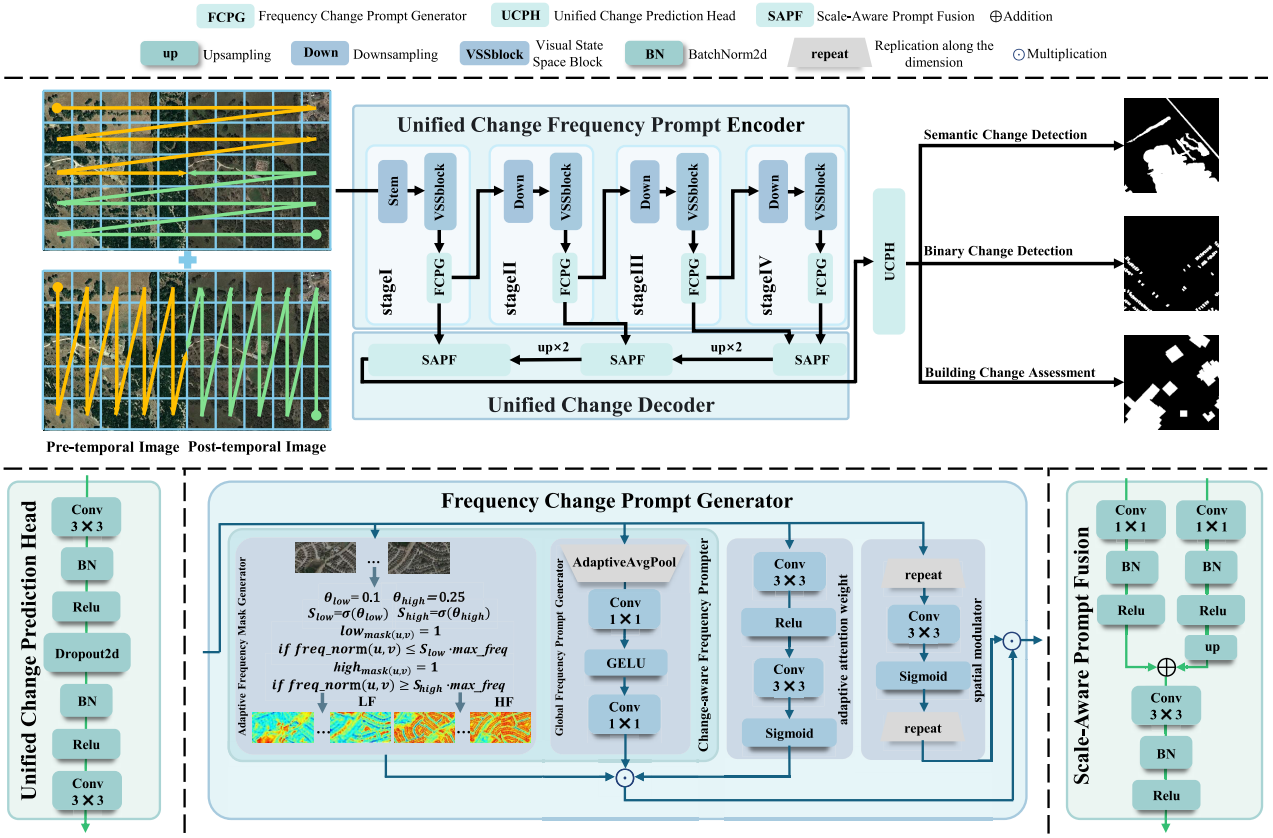


Fig. 2. Overall architecture of UniRSCD. The upper half of the figure shows the main workflow of UniRSCD. The horizontally concatenated bitemporal images are scanned in both forward and backward directions along rows and columns to serve as input. Data flows sequentially to the right through processing units from Stage I to Stage IV, which include FCPG and VSSBlock modules. After passing through the scale-aware prompt fusion (SAPF) module and undergoing upsampling, the data are directed via UCPH to three parallel task branches: SCD, BCD, and BDA. The lower half of the figure shows the internal structure of the FCPG, UCPH, and SAPF.

where α is an adjustable fusion coefficient. This design integrates global statistics, local frequency bands, and spatial weights, effectively fusing multidimensional information. By explicitly modeling in the frequency domain, the FCPG module robustly integrates global context with local details by obtaining global information and local information, enhancing detection performance in complex scenarios.

C. Unified Change Decoder and Prediction Head

As shown in Fig. 2, UniRSCD employs a simple feature pyramid network as its decoder and a unified prediction head for multiple CD tasks.

1) Unified Change Decoder: We design a unified change decoder based on a feature pyramid network to integrate multiscale encoder features F^1, F^2, F^3, F^4 via top-down propagation. First, features from each level are projected into a unified space using 1×1 convolutions, where P_i denotes the projected feature of the ith level. Then, a top-down fusion process starts from the highest level, progressively upsampling and merging features with lower levels as formulated by

$$N_i = Conv_{3\times3}\left(Up\left(N_{i+1}\right) \oplus P_i\right) \tag{4}$$

where Up denotes the bilinear upsampling operation and $\oplus$ represents elementwise addition. To enhance feature discriminability, skip connections with 3×3 convolutions are introduced at each fusion stage. The final change representation is then constructed by concatenating and refining the upsampled multiscale features, as $C = Conv_{3\times3}(Concat(Up(N_4), Up(N_3), Up(N_2), N_1))$, where Up denotes upsampling. The skip connections utilize 3×3 convolutions to align feature distributions, effectively aggregating the high-frequency edge details preserved by the FCPG with the rich semantic context from the decoder. This multiscale fusion strategy leverages both high-level semantics and low-level details, thereby enriching context for change prediction while improving boundary localization and semantic consistency. The reason for choosing a lightweight pyramid structure is that the proposed encoder captures both global and local information during the encoding process, making a complex decoder functionally redundant.

2) Unified Prediction Head: We design a unified prediction head that achieves unified support for various tasks, including BCD, SCD, and BDA, by task-specific adaptation. Given an input feature map $X \in R^{C_{in}\times H\times W}$, the unified feature extraction process can be formulated as

$$\begin{aligned} Z_1 &= Dropout\left(\sigma\left(BN\left(Conv_{3\times3}\left(X, W_1\right)\right)\right)\right) \\ Z_2 &= \sigma\left(BN\left(Conv_{3\times3}\left(Z_1, W_2\right)\right)\right) \end{aligned} \tag{5}$$

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where $\text{Conv}_{3\times3}$ is a 3×3 convolution, BN is batch normalization, σ is a configurable activation function (*ReLU*), and random feature dropping is applied with a probability $p = 0.2$, an empirical setting designed to prevent overfitting while preserving sufficient feature expressiveness. This design progressively reduces feature dimensions to $C_{hid}/2$, balancing capacity and complexity. The prediction heads for all tasks share an identical unified structure, with the main difference lying in the number of output branches and their specific output dimensions to accommodate different task requirements. In particular, for the multitask output: the BCD task utilizes a single prediction head with tailored output dimensions; the SCD task employs three parallel output heads for CD and bitemporal semantic segmentation, enhanced through background suppression mechanisms to better distinguish foreground from background; and the BDA task utilizes a dual-branch prediction head for simultaneous building segmentation and damage assessment. This unified framework supports multiple CD tasks, ensuring practicality and extensibility, as shown in Fig. 2.

D. Multitask Adaptive Loss Function

During the training of CD models, the appropriate loss function is automatically selected based on different task types. These loss functions are designed as a combination of cross-entropy loss [51] and Lovasz loss [16], ensuring both category balance optimization and optimization for the intersection over union (IoU) metric.

For BCD, a loss function formulation is designed to supervise the CD task. A combination of cross-entropy loss and Lovasz loss is adopted to handle pixel-level accuracy and segmentation quality. The final loss is defined as

$$L_{bcd} = L_{ce} + 0.75 \times L_{lovasz} \tag{6}$$

where L_{ce} provides category-balanced optimization, while L_{lovasz} directly optimizes the IoU metric. Due to the severe class imbalance in CD tasks, we assign a weight of 0.75 to the Lovasz loss in order to optimize the IoU metric more directly and enhance boundary quality.

For BDA, we extend the formulation (6) to simultaneously optimize building localization and damage classification. The multitask loss integrates

$$L_{bda} = L_{ce}^{loc} + L_{ce}^{clf} + 0.5 \times L_{lovasz}^{loc} + L_{lovasz}^{clf}. \tag{7}$$

L_{ce}^{loc} and L_{ce}^{clf} provide category-balanced supervision for building localization and damage classification, respectively, while L_{lovasz}^{loc} and L_{lovasz}^{clf} directly optimize the IoU metrics for localization and classification.

For SCD, we introduce a complex triple-branch supervision scheme, which includes CD, bitemporal semantic segmentation, and temporal consistency. The loss function comprises

$$L_{scd} = L_{ce}^{cd} + 0.5 \times \left(L_{ce}^{t1} + L_{ce}^{t2} + 0.5 \times L_{sim}\right) + 0.75 \times \left(L_{lovasz}^{cd} + 0.5 \times \left(L_{lovasz}^{t1} + L_{lovasz}^{t2}\right)\right). \tag{8}$$

In this formulation, L_{ce}^{cd}, L_{ce}^{t1}, and L_{ce}^{t2} provide category-balanced supervision for CD, semantic segmentation at image T_1, and semantic segmentation at image T_2, respectively, while L_{lovasz}^{cd}, L_{lovasz}^{t1}, and L_{lovasz}^{t2} optimize the corresponding IoU metrics for each task.

IV. Experiments and Results

A. Cd Task Descriptions

1) Binary CD: In the field of remote sensing CD, BCD serves as a fundamental task aimed at identifying and locating specific areas on the Earth's surface that have undergone alterations across different time periods. This task involves processing image pairs captured of the same area at distinct time points to generate a binary change mask, wherein each pixel is classified as either changed or unchanged. This approach focuses solely on whether change has occurred, without addressing the specific attributes of the change.

2) Semantic CD: As an extension of BCD, SCD not only identifies where changes occur but also determines their nature. It requires not only delineating the spatial extent of changes but also precisely classifying land-cover types within altered areas across different time periods. This provides comprehensive semantic information on which land-cover types transform into which others. Such refined change analysis is crucial for deepening our understanding of the dynamic evolution of land cover and land use.

3) Building Damage Assessment: In disaster emergency response, BDA represents a critical application scenario for SCD. This task focuses on individual building structures, requiring the integrated use of predisaster and postdisaster remote sensing imagery. It involves first precisely locating buildings, then evaluating their damage severity during the disaster. The final output comprises spatial distribution maps of buildings alongside corresponding damage grades, providing objective, quantified spatial information support for rapid disaster assessment and rescue decision-making.

B. Dataset Description

1) BCD Datasets:

1) The LEVIR-CD dataset comprises 637 sets of very high-resolution images, each with a resolution of 0.5 m per pixel and dimensions of 1024×1024 pixels. These bitemporal images, spanning 5 to 14 years, reveal significant land use changes. They also encompass diverse building types, including detached houses, high-rise flats, large warehouses, and small garages, with a particular focus on the emergence of new constructions and the decay of existing structures.
2) The WHU-CD dataset comprises imagery covering over 450 square kilometers across Wellington and Christchurch in New Zealand, featuring a spatial resolution of 0.3 m. It contains 8188 optical remote sensing images cropped to 512×512 pixels alongside corresponding annotated maps. Captured in April 2012 and 2016, respectively, these images document significant urban development occurring over a four-year period.
3) The SYSU-CD dataset comprises 20 000 sets of 0.5-m resolution aerial photographs, each measuring 256×256 pixels, captured across Hong Kong between 2007 and 2014. Primary change types include: new

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urban construction, suburban expansion, preconstruction groundworks, vegetation alterations, road widening, and offshore developments.

2) SCD Dataset: The SECOND dataset comprises 4662 sets of aerial imagery with resolutions ranging from 0.5 to 3 m per pixel, captured by multiplatform sensors across cities including Hangzhou, Chengdu, and Shanghai. This dataset encompasses six primary land-cover types: bare ground, trees, low vegetation, water bodies, buildings, and recreational areas. These zones are susceptible to geographical changes triggered by both natural and human factors.

3) BDA Dataset: The xBD dataset focuses on BDA in disaster response, integrating predisaster and postdisaster satellite imagery. Covering an area of 45 362 square kilometers, it contains annotated information for over 850 000 buildings. Adhering to rigorous standards, the dataset employs high-resolution imagery exceeding 0.8 m per pixel, categorizing damage into four levels: "No damage," "Minor damage," "Major damage," and "Destroyed." Applicable disaster types encompass earthquakes, tsunamis, storms, wildfires, volcanic eruptions, and floods, among others.

C. Experimental Setup

1) Datasets and Evaluation: We conduct a comprehensive evaluation on five public CD benchmark datasets: LEVIR-CD [4], WHU-CD [52], SYSU [53], SECOND [5], and xBD [6]. LEVIR-CD, WHU-CD, and SYSU are three BCD datasets, containing changes in buildings, roads, and lakes. SECOND is an SCD dataset with six land-cover categories for semantic learning. xBD is a BDA dataset providing information on four damage levels. All datasets use their official default splits for training, validation, and test sets. This ensures a fair comparison with other current SOTA methods. For the CD task, we adopt precision, recall, $F1$ score, and IoU; for SCD, we use overall accuracy (OA), mean IoU (mIoU), semantic Kappa coefficient (SeK), and the $F1$ score; and for disaster assessment, the localization $F1$ score, damage classification $F1$ score, and overall $F1$ score are employed [40].

2) Implementation Details: In this study, all experiments were conducted on a hardware platform equipped with an NVIDIA RTX 5880 Ada GPU, utilizing the PyTorch 2.0 framework and CUDA 11.8 for accelerated computing. To achieve the optimal balance between parameter efficiency and feature expressiveness, the proposed unified architecture employs the VMamba's small version [42]. For the BCD task, the training parameters were uniformly set as follows: batch size of 22, input image crop size of 256×256, maximum iterations of 320 000, initial learning rate of $1e^{-4}$, and weight decay of $5e^{-4}$. For the BDA task, the training parameters were set to: batch size of 16, input image crop size of 256×256, maximum iterations of 400 000, initial learning rate of $1e^{-4}$, and weight decay of $5e^{-4}$. For the SCD task, the parameters were: batch size of 16, input image crop size of 256×256, maximum iterations of 400 000, initial learning rate of $1e^{-4}$, and weight decay of $5e^{-4}$. Random rotation, left–right flipping, and top–bottom flipping were used as training data augmentation methods.

TABLE I

PERFORMANCE COMPARISON OF DIFFERENT BCD METHODS ON THE SYSU DATASET. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY. ALL RESULTS FOR THE THREE EVALUATION METRICS ARE EXPRESSED AS PERCENTAGES (%)

Method	Rec	Pre	F1	IoU	Params
BIT [25]	75.56	83.59	79.44	65.94	3.55M
SEIFNet [9]	78.35	83.59	80.88	67.9	39.08M
ChangeFormer [8]	78.51	77.16	77.83	63.71	41.03M
AFCF3D-Net [12]	83.88	82.3	83.11	71.09	17.54M
AACRNet [55]	82.62	84.55	83.57	71.78	13.50M
ChangeMamba [17]	77.82	87.89	82.55	70.28	49.94M
UniRSCD	80.62	87.11	83.74	72.03	50.27M

For all three CD tasks, this study employed a unified two-stage training strategy, with the first and second stages accounting for 90% and 10% of the total iterations, respectively. Taking the BCD task as an example, comprising a total of 320 000 iterations, these two stages consisted of 288 000 and 32 000 iterations, respectively. In each stage, a cosine-annealing learning rate scheduler was used to reduce the learning rate to $0.01\times$ its initial value for that stage. Upon entering the second stage, the initial learning rate was set to $1e^{-5}$. In the first stage, the FCPG does not participate in training and remains frozen. All other modules are trained using full learning rates and weight decay to learn robust spatiotemporal features. In the second stage, the backbone network is frozen, and only the FCPG and prediction head are activated and trained using a lower learning rate to refine details. For FCPG, convolutional layers are initialized via the Kaiming normal method, while biases and the SPM are zero-initialized. This prevents catastrophic forgetting and enables the FCPG to focus on refining the boundary between high and low frequencies (HLFs) within a stable feature space. The optimization process employs the AdamW optimizer [54].

D. Comparison With SOTA Methods

To validate the effectiveness of the proposed method, comprehensive comparisons are conducted with various representative SOTA methods in the field of remote sensing CD in recent years, including BIT [25], SEIFNet [9], ChangeFormer [8], AFCF3D-Net [12], AACRNet [55], ChangeMamba [17], HRSCD-S4 [56], ChangeMask [51], SSCD [14], BiSRNet [14], TED [13], xBD baseline [6], MTF [57], ChangeOS-R50 [16], ChangeOS-R101 [16], and DamFormer [58]. All comparative methods are reproduced based on their official codebases or directly cite the best reported results.

1) Binary CD: UniRSCD achieves an $F1$ score 0.63% higher than AFCF3D-Net on the SYSU dataset as illustrated in Table I. On the LEVIR-CD dataset, as illustrated in Table II, its $F1$ and IoU surpass ChangeMamba-Small by 0.09% and 0.15%, respectively. On the WHU-CD dataset, as illustrated in Table III, it achieved comparable performance to SOTA methods, demonstrating excellent general detection capabilities. Although UniRSCD's parameters are not exactly lightweight, its single-stream architecture avoids the significant architectural redundancy associated with dual-branch feature

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT BCD METHODS ON THE LEVIR-CD DATASET

Method	Rec	Pre	F1	IoU	Params
BIT [25]	89.37	89.24	89.31	80.68	3.55M
SEIFNet [9]	88.28	91.74	89.98	81.78	39.08M
ChangeFormer [8]	88.8	92.05	90.4	82.48	41.03M
AFCF3D-Net [12]	90.17	91.35	90.76	83.08	17.54M
AACRNet [55]	91.68	92.69	92.18	85.5	13.50M
ChangeMamba [17]	91.67	92.72	92.19	85.52	49.94M
UniRSCD	92.87	91.78	92.28	85.67	50.27M

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT BCD METHODS ON THE WHU-CD DATASET

Method	Rec	Pre	F1	IoU	Params
BIT [25]	81.48	86.64	83.98	72.39	3.55M
SEIFNet [9]	72.55	87.73	79.42	65.87	39.08M
ChangeFormer [8]	90.28	89.36	89.82	81.6	41.03M
AFCF3D-Net [12]	93.69	93.47	93.58	87.93	17.54M
AACRNet [55]	93.31	95.23	94.26	89.14	13.50M
ChangeMamba [17]	92.29	95.9	94.06	88.79	49.94M
UniRSCD	94.08	95.55	94.81	90.14	50.27M

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT SCD METHODS ON THE SECOND DATASET

Method	OA	F1	mIoU	SeK	Params
HRSCD-S4 [56]	83.88	47.69	70.58	16.81	13.71M
ChangeMask [51]	85.53	50.54	71.39	18.36	2.97M
SSCD [14]	85.62	52.8	72.14	20.15	23.31M
BiSRNet [14]	85.8	53.59	72.69	21.18	23.39M
TED [13]	85.5	52.61	72.19	20.29	24.19M
UniRSCD	86.44	56.35	72.7	22.56	54.43M

extraction, thereby demonstrating a significant advantage in actual training time.

2) Semantic CD: UniRSCD demonstrated exceptional performance on the SECOND dataset as illustrated in Table IV. UniRSCD surpassed ChangeMask by 5.81% in $F1$ score and 4.20% in semantic Kappa, achieving optimal results on most metrics and highlighting its superior capability in complex semantic understanding.

3) Building Damage Assessment: As illustrated in Table V, UniRSCD achieved an overall $F1$ score exceeding DamFormer by 1.00% on the xBD dataset, with a remarkable 85.22% precision for detecting "destroyed" damage levels, further validating its multitask adaptability.

These improvements demonstrate the effectiveness of the UniRSCD unified single-stream Mamba architecture and frequency change prompt module. The former effectively captures long-range spatiotemporal dependencies through a selective SSM, fully interacting with bitemporal features. The latter achieves a comprehensive feature representation by obtaining global information and local information by employing candidate region localization guided by low-frequency holistic information and boundary optimization driven by high-frequency detail information.

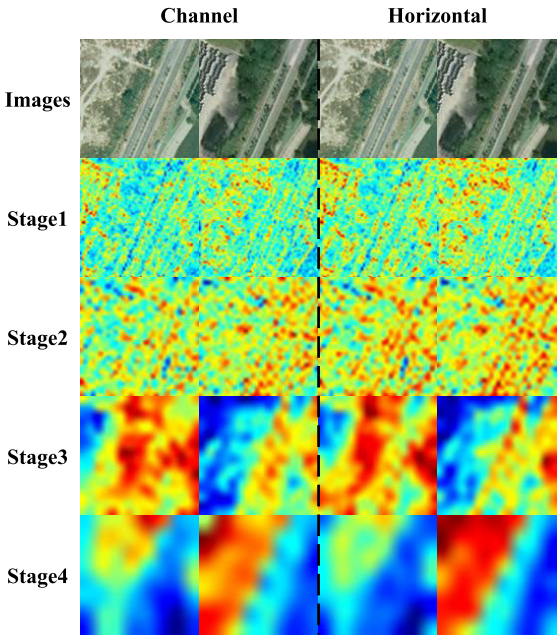


Fig. 3. Feature map comparison of bitemporal concatenation methods on the SYSU dataset.

E. Diagnostic Study

We selected the SYSU dataset as the benchmark for our ablation experiments because it requires comprehensively capturing global and local information, which can effectively validate the functionality of the core components, thereby systematically verifying the necessity and synergy of each core component within the UniRSCD framework. All experiments employed identical training settings to ensure fair comparisons.

1) Bitemporal Concatenation Methods: To validate the effectiveness of the bitemporal horizontal concatenation method, we conducted relevant experiments comparing the horizontal concatenation approach with the channel concatenation approach. We have removed the FCPG from UniRSCD as an experimental model, named UniRS-noFCPG. Tables VII, XIV, and XV compare the performance of channel concatenation versus horizontal concatenation within the UniRS-noFCPG framework across BCD, SCD, and BDA tasks. The results clearly demonstrate the superiority of lateral concatenation, which exhibits both higher efficiency and greater stability. Unlike channel concatenation, horizontal concatenation connects two feature maps side-by-side, enabling the framework to learn correlations between corresponding pixels across the bitemporal feature maps. This approach captures changing signals more directly and accurately. Figs. 3–5 show the feature visualizations of the encoder.

2) Training Time Comparison: We compared the runtime of each method on the SYSU dataset under identical hardware conditions. Training time exceeds that of current SOTA methods such as ChangeMamba [17]. As illustrated in Table XVI,

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TABLE V
PERFORMANCE COMPARISON OF DIFFERENT BDA METHODS ON THE XBD DATASET

Method	F_1^{loc}	F_1^{clf}	$F_1^{overall}$	Damage F_1 per-class				Params
				No damage	Minor damage	Major damage	Destroyed	
xBD baseline [6]	80.47	3.42	26.54	66.31	14.35	0.94	46.57	25.72M
MTF [57]	83.6	70.02	74.1	90.6	49.3	72.2	83.7	44.40M
ChangeOS-R50 [16]	85.41	70.88	75.24	88.98	53.33	71.24	80.6	39.05M
ChangeOS-R101 [16]	85.69	71.14	75.5	89.11	53.11	72.44	80.79	58.05M
DamFormer [58]	86.86	72.81	77.02	89.86	56.78	72.56	80.51	32.52M
UniRSCD	84.47	75.26	78.02	94.76	56.64	75.71	85.22	52.44M

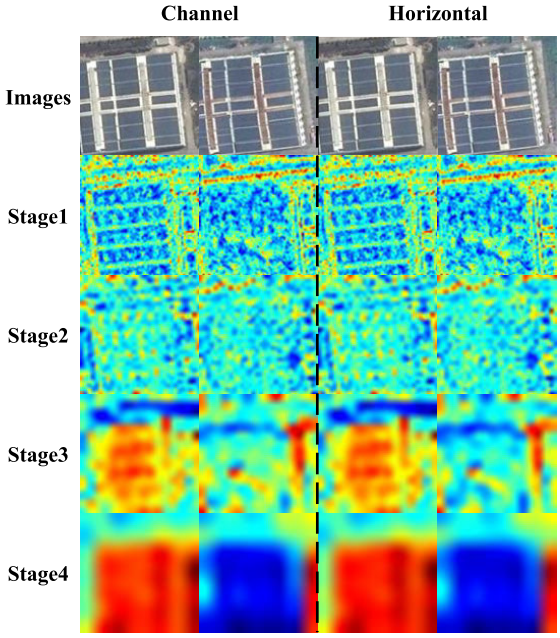


Fig. 4. Feature map comparison of bitemporal concatenation methods on the SECOND dataset.

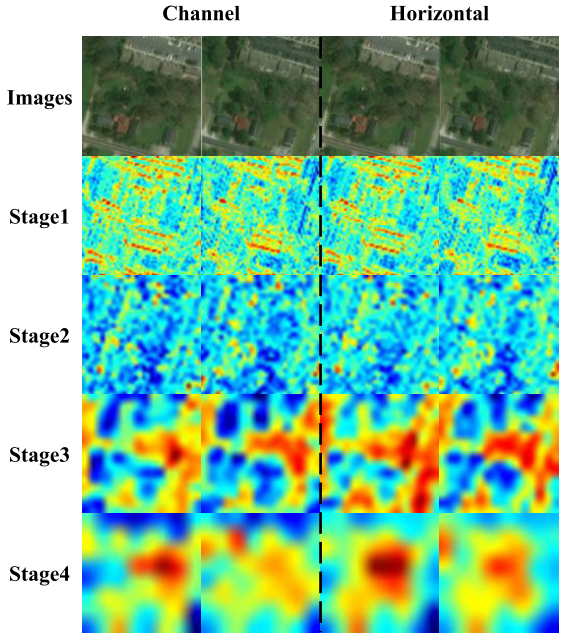


Fig. 5. Feature map comparison of bitemporal concatenation methods on the xBD dataset.

TABLE VI
VALIDATION OF THE EFFECTIVENESS OF DIFFERENT MODULES OF UNIRSCD ON THE SYSU DATASET. "PRM" IS PRETRAINED MODEL, "FCPG" IS FCPG, "AAF" IS ADAPTIVE ADJUSTMENT OF FREQUENCY SETTINGS, "SPM" IS SPM, AND "HLF" IS HLFS

PrM	FCPG	SPM	HLF	AAF	Rec	Pre	F1	IoU
	✓	✓	✓	✓	76.56	80.09	78.29	71.79
✓					79.11	86.78	83.42	71.56
✓	✓				82.45	85.23	83.07	71.52
✓	✓	✓			82.31	85.74	83.14	71.67
✓	✓	✓	✓		79.76	86.54	83.16	71.23
✓	✓	✓	✓	✓	80.62	87.11	83.74	72.03

the proposed UniRSCD algorithm demonstrated superior computational speed.

3) Frequency Prompt Change Modeling Mechanism: We first evaluated the contribution of the overall architecture design to performance, as illustrated in Table VI. Experimental results demonstrate that incorporating the FCPG module improves performance across all tasks, achieving an $F1$ score improvement of 0.32% and an IoU improvement of 0.47%. To further validate the effectiveness of the frequency prompt modeling mechanism, we conducted an in-depth ablation analysis of the internal components of the FCPG. Experimental results indicate that removing the SPM component resulted in a 0.51% decrease in Prescore. This validates that the component effectively enhances the model's ability to capture fine-grained structures by improving the sensitivity of boundary region responses through spatial adaptive weighting within local receptive fields. Regarding frequency settings, replacing adaptive adjustment of high- and low-frequency settings (AAF) with fixed thresholds (low-frequency 0.1, high-frequency 0.25) resulted in a 0.57% decrease in Prescore while replacing both HLFs with a single frequency threshold (0.1) caused a 1.2% drop in Prescore. This clearly demonstrates that the adaptive high–low frequency coordination mechanism not only achieves coarse localization of change regions through low-frequency information but also refines boundaries using high-frequency information. They form a detection process progressing from coarse to fine, jointly ensuring the final detection accuracy. At the same time, because datasets such as LEVIR-CD and SECOND have significantly different spatial

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TABLE VII
COMPARISON OF DIFFERENT CONCATENATION METHODS ON THE XBD DATASET

Bitemporal concatenation method	F_1^{loc}	F_1^{clf}	$F_1^{overall}$	Damage F_1 per-class			
				No damage	Minor damage	Major damage	Destroyed
Channel concatenation	79.85	74.47	76.08	**94.89**	57.33	72.76	83.39
Horizontal concatenation	**83.95**	**75.55**	**78.07**	94.42	**57.80**	**73.89**	**86.80**

TABLE VIII
COMPARISON OF DIFFERENT ENCODER NETWORKS UNDER SINGLE-STREAM ARCHITECTURE ON THE SYSU DATASET. UCD DENOTES THE UNIFIED CHANGE DECODER, AND UCPH DENOTES THE UCPH

Method	Rec	Pre	F1	IoU
ResNet-50 + UCD + UCPH	80.18	85.84	82.91	**70.82**
ViT-Base + UCD + UCPH	78.35	84.00	81.08	68.18
Mamba + UCD + UCPH	**80.95**	**86.33**	**83.13**	70.68

TABLE IX
COMPARISON OF DIFFERENT ENCODER NETWORKS UNDER SINGLE-STREAM ARCHITECTURE ON THE SECOND DATASET

Method	OA	F1	mIoU	SeK
ResNet-50 + UCD + UCPH	84.17	45.91	71.26	15.72
ViT-Base + UCD + UCPH	84.84	49.17	68.48	15.61
Mamba + UCD + UCPH	**86.38**	**56.14**	**72.60**	**22.48**

TABLE X
COMPARISON OF DIFFERENT ENCODER NETWORKS UNDER THE DUAL-STREAM ARCHITECTURE ON THE SYSU DATASET

Method	Rec	Pre	F1	IoU
ResNet-50 + UCD + UCPH	**80.78**	**84.18**	**82.45**	70.14
ViT-Base + UCD + UCPH	68.74	77.55	72.88	57.33
Mamba + UCD + UCPH	80.31	83.00	82.33	**70.34**

TABLE XI
COMPARISON OF DIFFERENT ENCODER NETWORKS UNDER THE DUAL-STREAM ARCHITECTURE ON THE SECOND DATASET

Method	OA	F1	mIoU	SeK
ResNet-50 + UCD + UCPH	84.12	39.70	69.66	10.78
ViT-Base + UCD + UCPH	84.84	49.17	68.48	15.61
Mamba + UCD + UCPH	**86.18**	**55.45**	**72.87**	**22.32**

resolutions, the AAF mechanism allows the model to dynamically adjust frequency-domain thresholds to match the actual scales of ground objects in different datasets, thereby ensuring robustness across different resolutions. Ablation experiments demonstrate that the FCPG modeling mechanism ensures precise recognition of local detail features and effective acquisition of global information and local information through the synergistic interaction of two core components, i.e., adaptive frequency of high- and low-frequency settings and SPM.

4) Feature Maps Visualization: We randomly sampled data from the SYSU, LEVIR-CD, and WHU-CD datasets and performed visual analysis on the feature maps processed by the frequency change prompt module across the encoder's four stages as shown in Fig. 6.

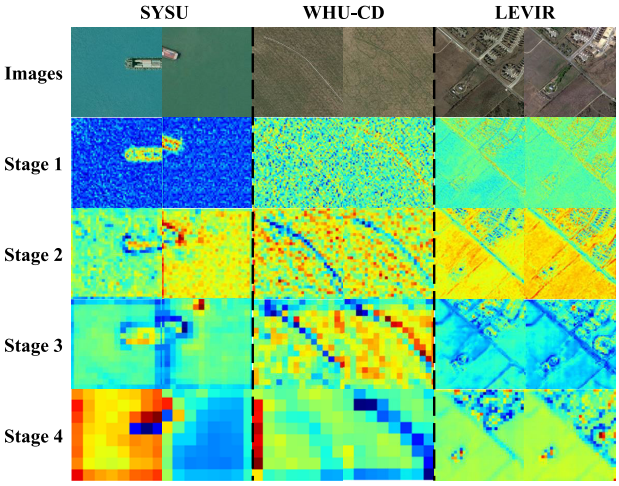


Fig. 6. Visualization of features at different layers of UniRSCD across three datasets.

F. Dual-Stream and Single-Stream Architectures Effectiveness Analysis

In the field of remote sensing CD, single-stream [59] and dual-stream [20] architectures represent two core feature processing paradigms within deep learning. We here conduct an effectiveness analysis of these two integration approaches.

1) Single-Stream Architecture Analysis: To investigate which approach is most suitable for the encoder within a unified CD framework, we conducted experiments using Mamba [17], ResNet-50 [60], and ViT-Base-Patch16-224 [61], respectively, all based on a single-stream architecture. These three baselines are selected as they represent the three dominant paradigms in computer vision: CNN, transformer, and SSM, providing a comprehensive and representative architectural comparison. This experiment exclusively utilized Mamba, ResNet-50, and ViT-Base-Patch16-224 as encoders, with the encoders not incorporating our proposed FCPG. The experiments employed our proposed unified change decoder and unified change prediction head (UCPH). As illustrated in Tables VIII, IX, and XII, the SSM demonstrates outstanding performance when embedded as the encoder network in the unified architecture across BCD, SCD, and BDA tasks. This indicates its significant advantage in the collaborative representation of bitemporal remote sensing imagery. As shown in Figs. 7 and 8, the encoder feature visualizations are presented

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TABLE XII

COMPARISON OF DIFFERENT ENCODER NETWORKS UNDER SINGLE-STREAM ARCHITECTURE ON THE XBD DATASET

Method	F_1^{loc}	F_1^{clf}	$F_1^{overall}$	Damage F_1 per-class			
				No damage	Minor damage	Major damage	Destroyed
ResNet-50 + UCD + UCPH	79.67	69.72	72.71	94.11	48.58	72.13	81.33
ViT-Base + UCD + UCPH	38.40	27.79	30.97	87.76	23.53	23.50	21.06
Mamba + UCD + UCPH	**83.95**	**75.55**	**78.07**	**94.42**	**57.80**	**73.89**	**86.80**

TABLE XIII

COMPARISON OF DIFFERENT ENCODER NETWORKS UNDER THE DUAL-STREAM ARCHITECTURE ON THE XBD DATASET

Method	F_1^{loc}	F_1^{clf}	$F_1^{overall}$	Damage F_1 per-class			
				No damage	Minor damage	Major damage	Destroyed
ResNet-50 + UCD + UCPH	82.26	72.72	75.85	94.21	52.60	72.69	86.04
ViT-Base + UCD + UCPH	38.39	46.46	44.04	82.09	32.77	46.81	45.13
Mamba + UCD + UCPH	**84.24**	**75.02**	**77.89**	**94.27**	**56.55**	**73.61**	**86.37**

TABLE XIV

COMPARISON OF DIFFERENT CONCATENATION METHODS ON THE SYSU DATASET

Bitemporal concatenation method	Rec	Pre	F1	IoU
Channel concatenation	**80.95**	86.33	83.13	70.68
Horizontal concatenation	79.11	**86.78**	**83.42**	**71.56**

TABLE XV

COMPARISON OF DIFFERENT CONCATENATION METHODS ON THE SECOND DATASET

Bitemporal concatenation method	OA	F1	mIoU	SeK
Channel concatenation	84.54	50.16	69.99	17.00
Horizontal concatenation	**86.38**	**56.14**	**72.60**	**22.48**

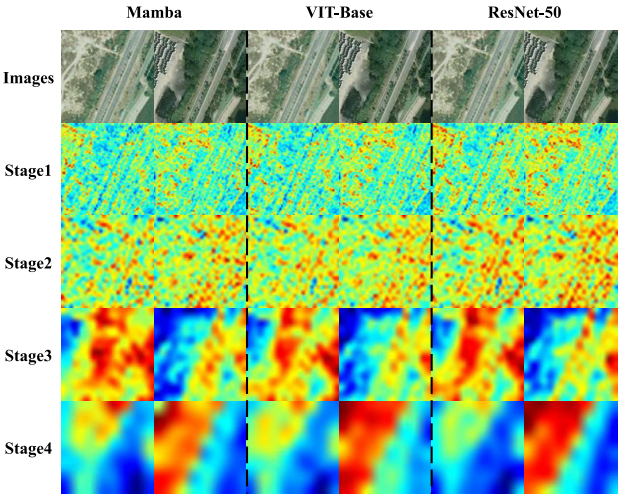


Fig. 7. Feature map comparison of three encoders under the single-stream architecture on the SYSU dataset.

for Mamba, ResNet-50, and ViT-Base-Patch16-224 under the single-stream architecture.

2) Dual-Stream Architecture Analysis: To investigate which approach is most suitable for the encoder within a unified

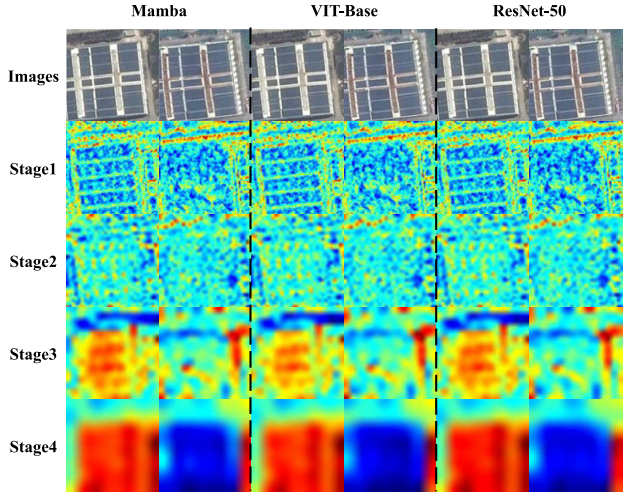


Fig. 8. Feature map comparison of three encoders under the single-stream architecture on the SECOND dataset.

TABLE XVI

RUNTIME COMPARISON ON THE SYSU DATASET. H DENOTES HOUR

Method	UniRSCD	ChangeMamba
Training time	4h	7h

CD framework, we conducted experiments using Mamba, ResNet-50, and ViT-Base-Patch16-224, respectively, all based on a dual-stream architecture. This experiment exclusively utilized Mamba, ResNet-50, and ViT-Base-Patch16-224 as encoders, with the encoders not incorporating the proposed FCPG. The experiments employed our proposed unified change decoder and UCPH. In comparative experiments on dual-stream architectures, as illustrated in Table X, although ResNet-50 outperformed Mamba on three metrics within the SYSU dataset, the advantage was marginal. Conversely, Mamba demonstrated substantial superiority over CNNs in the other two tasks tested. As demonstrated by Tables XI and XIII, SSMs exhibited formidable competitiveness. By independently

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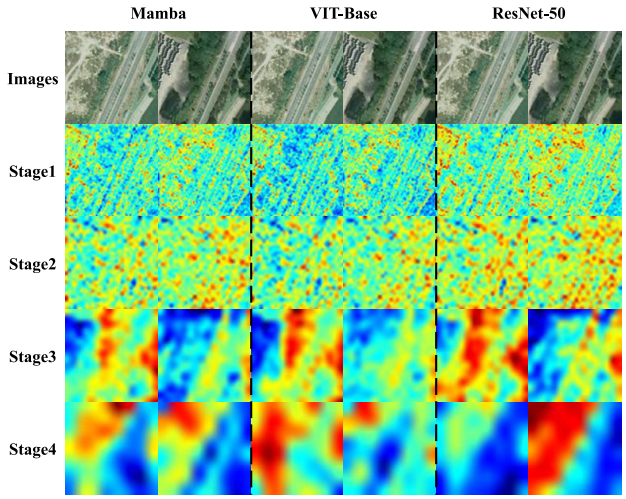


Fig. 9. Feature map comparison of three encoders under the dual-stream architecture on the SYSU dataset.

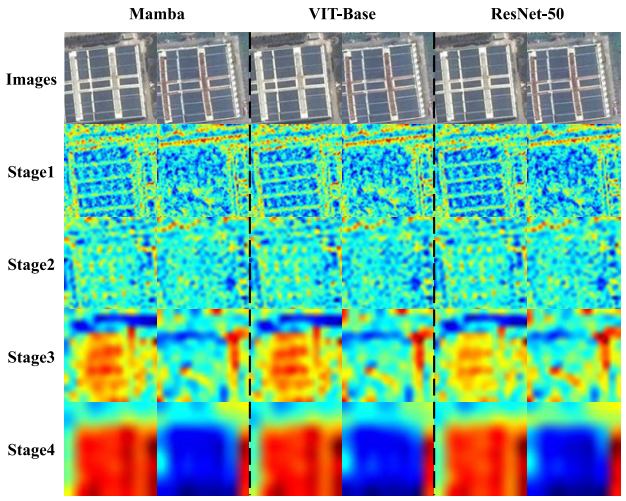


Fig. 10. Feature map comparison of three encoders under the dual-stream architecture on the SECOND dataset.

processing bitemporal images and performing deep feature comparisons, this architecture demonstrates unique value in BDA. As shown in Figs. 9 and 10, the encoder feature visualizations are presented for Mamba, ResNet-50, and ViT-Base-Patch16-224 under the dual-stream architecture.

G. Visualization of Algorithm Results

1) Binary CD: Fig. 11 shows the binary change maps predicted by our UniRSCD across the SYSU dataset. It can be observed that the selected architecture more accurately detects the various alterations present within these image pairs. The test results exhibit a high degree of correspondence with the change reference maps. In particular, as highlighted by the red dashed boxes in the first row, in challenging areas characterized by complex natural terrain and vegetation cover, UniRSCD preserves sharp and continuous object boundaries. The visual results show that UniRSCD clearly detects the irregular outlines and boundaries of natural landcover changes.

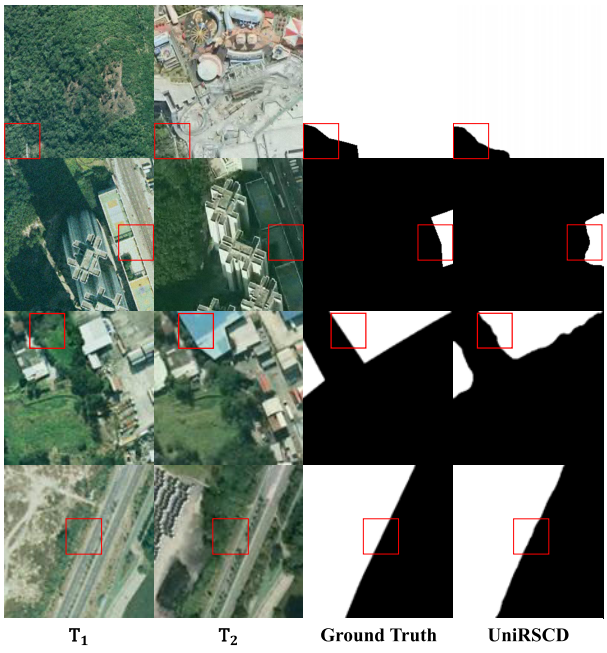


Fig. 11. Several binary change maps obtained via our method on the test set of the SYSU dataset. The red dashed boxes highlight the challenging areas characterized by complex natural terrain and vegetation cover.

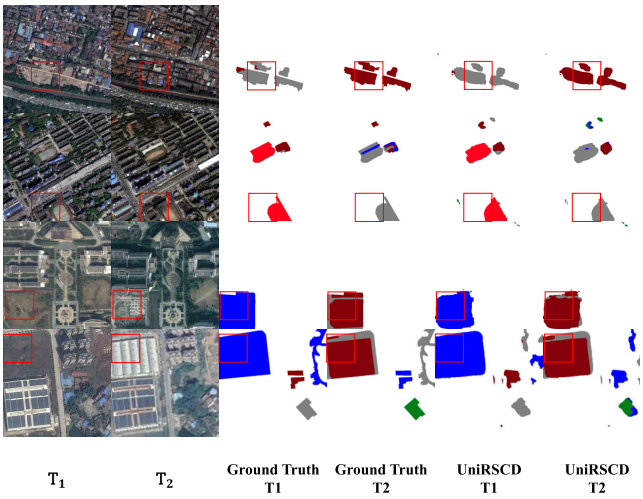


Fig. 12. Some semantic change maps obtained via the proposed UniRSCD on the SECOND dataset's test set. White indicates unchanged areas; blue denotes water bodies; medium gray represents bare ground or natural vegetation; dark green signifies low vegetation; bright green indicates trees; dark red denotes buildings; and light red represents playground areas.

2) Semantic CD: Fig. 12 shows some semantic change maps on the SECOND dataset test set. It can be observed that UniRSCD not only distinguishes variation features across different scales but also accurately identifies the specific semantic categories of these variations. As emphasized by the red dashed boxes in the first row, the model maintains high robustness and precision in complex urban environments with dense structures. It successfully avoids boundary confusion among adjacent land-cover types. In particular, our method

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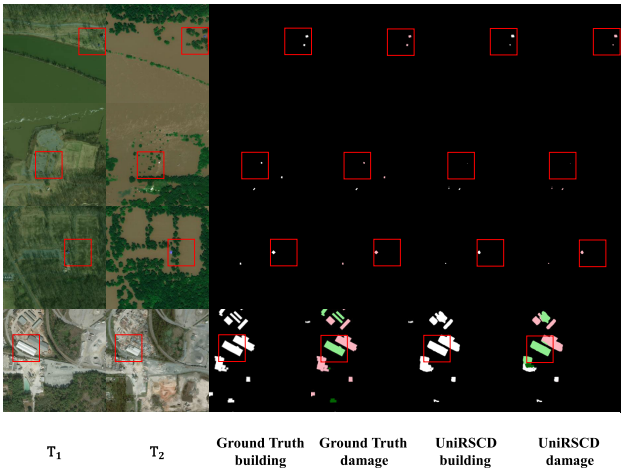


Fig. 13. Some BDA results obtained using our method on the test set of the xBD dataset. Background is marked in black; building is indicated in white; destroyed is marked in dark green; no damage is annotated in light green; minor damage is indicated in light red; and major damage is displayed in dark red.

keeps large uniform areas complete while maintaining clear edges where different land types meet.

3) Building Damage Assessment: Fig. 13 shows qualitative predictions on the xBD dataset, illustrating UniRSCD's capability to delineate building footprints and categorize damage severities across diverse building structures and disaster scenarios. Furthermore, as illustrated in the red dashed boxes in the fourth row, for building regions or structures, UniRSCD precisely delineates individual building footprints and accurately distinguishes subtle damage grades, confirming its effectiveness in capturing fine details under extreme disaster conditions.

V. Conclusion

A novel unified CD paradigm named UniRSCD has been proposed here. The framework integrates SSMs with frequency change prompts into a unified encoder, enabling the model to simultaneously capture both global and local information. This effectively addresses the information loss inherent in traditional encoding processes. Furthermore, a structurally simple decoder alongside a multitask adaptive prediction head is introduced, integrating tasks with varying output granularities into a single architecture. Comprehensive experiments validate the generality of the proposed UniRSCD, showing that it achieves SOTA results on a diverse set of CD tasks. It is our hope that our work will serve diverse CD tasks and inspire further research in related fields.

Acknowledgment

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